

CITY UNIVERSITY OF HONG KONG

Comparative Air Pollution Analysis:
Mexico City vs. Hong Kong

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ABSTRACT

Megacities such as Mexico City and Hong Kong face recurrent air-pollution episodes shaped by their contrasting geography and meteorology. Mexico City lies in a high-altitude, enclosed basin where weak winds and frequent inversions trap locally produced pollutants, while Hong Kong is a coastal, island-rich region where pollution episodes are strongly influenced by monsoon flows, regional transport, and tropical-cyclone modulation. Understanding the meteorological drivers of extreme pollution is therefore essential for designing effective air-quality and public-health strategies.

This guided study examines the large-scale circulation pattern associated with high-pollution days in Mexico City and compares it conceptually with Hong Kong's documented regimes and health-based air-quality indices. Using 2012–2024 surface observations ($\text{PM}_{2.5}$, PM_{10} , O_3 , NO_2 , SO_2) and daily NARR pressure-level fields, “event days” were identified from monthly 90th-percentile thresholds, and 500-hPa geopotential-height anomaly composites were computed relative to a smoothed day-of-year climatology.

Across all pollutants, Mexico City's event days consistently show a mid-tropospheric ridge with positive 500-hPa anomalies (+4 to +6 m) and very weak ENE flow ($< 2 \text{ m s}^{-1}$) over the basin, indicating a clear ridge–stagnation regime that favors pollutant accumulation near the surface. Seasonal cycles differ among pollutants: O_3 peaks in April–May, PM_{10} and NO_2 reach their highest values in winter and early spring, and $\text{PM}_{2.5}$ shows elevated concentrations in both winter and late spring, while SO_2 remains low year-round. All pollutants exhibit minima during the rainy season (June–September), when enhanced convection and deeper mixing improve ventilation. These results agree with previous Mexican studies on blocking highs, basin-scale stagnation, and boundary-layer suppression.

In contrast, Hong Kong literature highlights that high-pollution events typically arise from imported aerosols, monsoon-induced subsidence, or specific tropical-cyclone tracks. Both cities, therefore, exhibit similar physical ingredients (subsidence and weak winds) but under different synoptic forcing and policy contexts. Recognizing these patterns can support early-warning tools and contribute to more targeted air-quality communication strategies.

1. INTRODUCTION

1.1 Urban air pollution in megacities

Air pollution is a major environmental health risk worldwide, and megacities typically combine dense populations, heavy traffic, industrial activity, and complex meteorology within a relatively small area. These conditions expose millions of people to multiple pollutants and create challenges for clear public communication and risk perception (Cromar et al., 2021).

Mexico City and Hong Kong are two contrasting yet comparable megacities that face persistent air-quality challenges. Although they lie at similar latitudes, they differ strongly in topography, emissions, atmospheric circulation, and regulatory systems, making them a meaningful pair for comparative analysis and for identifying insights that each city can learn from the other.

1.2 Mexico City basin, Hong Kong coastal setting

The Mexico City Metropolitan Area (MCMA) is situated in a high-altitude enclosed basin, roughly 2,300 meters above sea level, surrounded by mountains that reach over 4,500 meters and are part of the Trans-Mexican Volcanic Belt (Díaz-Esteban et al., 2022). Frequent morning thermal inversions in winter and spring, limited ventilation during the dry season, and a shallow boundary layer restrict the dispersion of locally generated pollutants (Burgos-Cuevas et al., 2021). Nevertheless, during the wet season, deeper mixing and rainfall reduce the frequency of extreme pollution events.

Hong Kong, on the other hand, is a subtropical coastal city shaped by a complex land-sea configuration. Air quality is modulated by monsoon flows, sea-land breezes, tropical-cyclone-related subsidence, and regional transport from the Pearl River Delta (Liu & Chan, 2002; Lu & Wang, 2004; Luo et al., 2018; Zeng et al., 2025). These processes create coastal recirculation and stagnant episodes that can lead to multi-pollutant peaks.

1.3 Research questions and objectives

Motivated by this contrast, yet recognizing some similarities to a certain extent, the main research question of this report is:

What large-scale circulation pattern governs extreme air-pollution days in Mexico City, and how does it compare with Hong Kong's regional meteorology and health-based air-quality indices?

To address this question, the guided study was organized around three nested objectives: (1) **to compare pollutant characteristics and health-based indices** between both cities (analyzed in *Weeks 1-3*); (2) **to analyze spatial and seasonal patterns in Mexico City**, focusing on differences between urban and rural stations (studied in *Weeks 2-3*); and (3) **to diagnose the synoptic-scale circulation during high-pollution days in Mexico City** and compare the results with documented Hong Kong meteorological regimes (performed in *Weeks 4-6*).

1.4 Scope and structure of the report

This study focuses on daily air-pollution data from 2012-2024 and the composite analysis of 500-hPa circulation over central Mexico. Due to time constraints and data-processing limitations, the composite analysis was performed only for Mexico City and only at one pressure level. However, this was sufficient to address the research question and identify a consistent synoptic fingerprint for extreme pollution events.

The report follows the six-week development of the guided study. *Section 2* reviews relevant literature on air pollution, meteorology, and health-based indices in Mexico City and Hong Kong. *Section 3* summarizes the weekly progression and study design. *Section 4* describes the datasets and composite-analysis methodology. *Section 5* presents the main results. *Section 6* discusses their implications, and *Section 7* outlines the main conclusions and future directions.

2. LITERATURE REVIEW

2.1 Health impacts of air pollution in Mexico City

Multiple epidemiological studies in Mexico City have quantified the health burden associated with air pollution, linking short-term exposure to increases in emergency department visits and long-term exposure to higher mortality (López-Feldman et al., 2021; Ugalde-Resano et al., 2022). Average annual PM_{2.5} levels in the MCMA typically range between 17 and 27 $\mu\text{g m}^{-3}$, and when combined with the region's high population density, an aging population, and a high prevalence of chronic diseases, overall vulnerability becomes substantial. These factors create a situation where even moderate increases in daily pollutant levels can lead to noticeable public health impacts, especially during periods of limited atmospheric ventilation.

For cardiovascular and respiratory emergencies, Ugalde-Resano et al. (2022) showed that a 10 $\mu\text{g m}^{-3}$ increase in PM₁₀ or PM_{2.5} is associated with roughly a 10% rise in emergency visits over short lags of a few days. This highlights the need for timely pollution alerts and for public-health services to anticipate spikes in acute respiratory and cardiac events during high-pollution episodes. Their findings also emphasize how quickly health outcomes can respond to changes in air quality, illustrating that even short-lived peaks in pollution can exert measurable pressure on hospital systems, especially in densely populated districts within the city.

At longer time scales, López-Feldman et al. (2021) found that chronic PM_{2.5} exposure across socially and spatially unequal areas of the MCMA significantly amplified COVID-19 mortality risk. Older adults and individuals with pre-existing conditions—such as diabetes, obesity, or hypertension—were particularly affected. As illustrated in **Figure 1**, communities in the central and northern parts of the MCMA experience consistently high PM_{2.5} concentrations due to nearby industrial corridors. These spatial inequalities not only shaped pandemic outcomes but also reflect broader patterns of environmental vulnerability, where neighborhoods with fewer resources face both higher exposures and greater health risks over time.

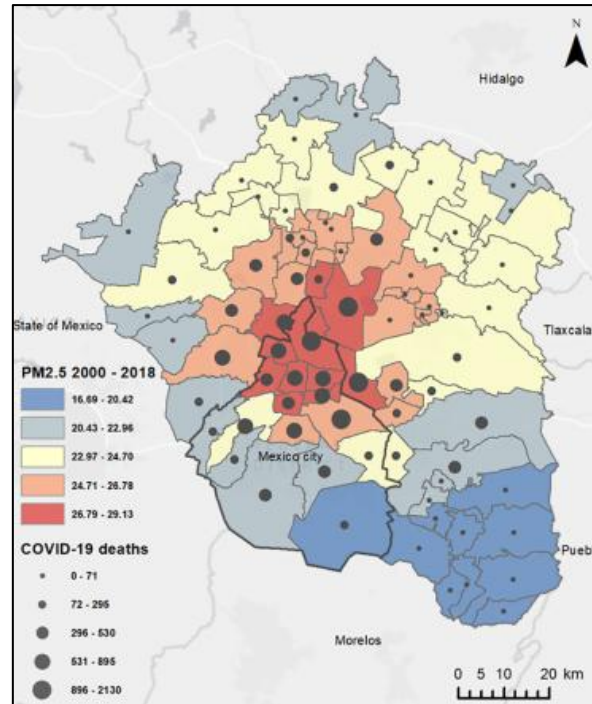


Figure 1. Distribution of long-term PM_{2.5} pollution and COVID-19 fatalities in Mexico City and the MCMA (López-Feldman et al., 2021).

Building on this health evidence, Cromar et al. (2021) proposed a multi-pollutant, health-based index for Mexico City, which was later adopted by the Mexico City Ministry of the Environment (SEDEMA) as “Know your Number.” This initiative became a key precursor to the current Air and Health Index (IAyS) (Aire CDMX, 2025b). Unlike concentration-based thresholds, this approach establishes a direct connection between pollutant levels and observed health risks, providing a significant epidemiological advantage that enhances public communication and informs more effective policy design.

2.2 Boundary-layer structure and local flows in the MCMA

From a meteorological standpoint, the Mexico City basin exhibits a highly structured atmospheric boundary layer (ABL), with frequent winter and spring inversions and a shallow mixed layer during the dry season, both of which favor pollutant accumulation (Burgos-Cuevas et al., 2021). During this season, the ABL is more stable and its early-morning depth is often minimal, matching the time of day when emissions peak. Only by the afternoon does the mixed layer grow enough to improve ventilation, which helps explain why relatively small variations in heating or overnight cooling can noticeably affect pollution levels.

Further studies demonstrated that local wind regimes and nocturnal convergence patterns significantly influence the distribution of pollutants within the basin (Salcido et al., 2019). Their work identified several recurrent surface-flow patterns that correspond to distinct pollutant distribution regimes across the MCMA. The combination of topographic confinement, nighttime convergence, and sensitivity to even slight shifts in wind direction means that weak synoptic forcing can have pronounced effects on surface air quality, particularly on stagnant days when boundary-layer growth is limited and air tends to remain trapped near the surface.

Additionally, research has found that the synergy between the urban heat island and the pollution island further contributes to spatial disparities in particulate matter (Mendez-Astudillo et al., 2022). Warmer urban surfaces and stable nocturnal conditions reinforce pollutant build-up in central “hotspots,” while surrounding rural areas tend to remain comparatively cleaner. This highlights the interaction between anthropogenic heat sources, the boundary-layer structure, and local emissions, which together influence the spatial variability of pollution across the Mexico City Metropolitan Area.

2.3 Synoptic patterns and dust transport over Mexico

Beyond boundary-layer processes, several studies have shown that synoptic-scale circulation and long-range transport strongly influence air quality across Mexico (Díaz-Esteban et al., 2022; Silva-Quiroz et al., 2019). Persistent subtropical highs, mid-tropospheric ridging, and prolonged dry periods suppress ventilation and intensify pollution episodes, especially during late winter and spring, providing the broader context in which local accumulation occurs.

Other studies demonstrated that Saharan dust intrusions during June-July can double or even quintuple PM concentrations in parts of Mexico, particularly in central and eastern states, therefore effectively elevating the background on which local emissions accumulate (Kuttralam-Muniasamy et al., 2021; Raga et al., 2021). Their findings also identified positive associations between PM levels and COVID-19 cases and deaths, reinforcing the importance of considering both short-term exposure and the chronic health burden during dust events.

Within the MCMA, Silva-Quiroz et al. (2019) linked ozone contingencies to 500-hPa blocking events, where high temperatures, weak winds, and in some cases Rossby-wave breaking

contribute to stagnation. Díaz-Esteban et al. (2022) further classified three decades of 500-hPa circulation into seven weather patterns, showing that high-pressure systems dominate O₃ peaks in spring, and PM_{2.5} and NO₂ peaks in winter, with La Niña years favoring a higher frequency of these ridge-like regimes. Their analysis found that such patterns produce the largest fraction of days exceeding WHO guidelines (World Health Organization, 2021), underscoring the direct role of synoptic circulation in shaping pollution extremes in the basin.

2.4 Health-based indices: IAYS, AQHI and WHO AQGs

In parallel with advances in epidemiological evidence, both Mexico City and Hong Kong have adopted updated health-based air-quality indices (AQIs) that aim to translate pollutant concentrations into clear, actionable categories. Unlike purely concentration-based systems, these indices incorporate health studies and exposure-response relationships to better reflect actual short-term risks for different population groups.

Mexico City's Air and Health Index (IAYS), formalized through norm NOM-172-SEMARNAT-2023, evolved from previous local indices and incorporates elements from international frameworks. It uses pollutant-specific thresholds, a 12-hour NowCast approach for particulate matter—like that used by the U.S. Environmental Protection Agency—and one-hour bands for O₃, NO₂, and SO₂ (SEMARNAT, 2023). The norm also establishes a gradual tightening of PM limits between 2023 and 2026. For ozone, the 8-hour metric was removed and replaced by the 1-hour average to better communicate acute risk, reduce confusion among the general population, and align with the 2021 WHO guidelines for short-term exposure. The index includes recommendations tailored to sensitive groups, reflecting its health-based orientation.

Hong Kong's Air Quality Health Index (AQHI) follows a different logic. Instead of evaluating pollutants individually, the AQHI combines O₃, NO₂, SO₂, and PM into a single composite value based on a 3-hour moving average. Each category—from “low” to “serious”—corresponds to estimated short-term hospital admission risks derived from local epidemiological studies (Environmental Protection Department, 2025). This approach allows the index to convey cumulative effects from multiple pollutants and to trigger health advisories when combinations of pollutants reach harmful levels.

Comparisons with the World Health Air Quality Guidelines (WHO AQGs) reveal that both local indices are less strict than WHO recommendations for annual and 24-hour limits. As shown in **Table 1**, WHO thresholds remain considerably lower than both IAYS and AQHI categories. However, Mexico City’s phased tightening brings its $PM_{2.5}$ and PM_{10} thresholds closer to WHO ranges, reflecting ongoing efforts to incorporate epidemiological evidence into regulatory design. Local indices ultimately respond to regional factors—emissions, weather patterns, and population health profiles—but also to local policies, which explain why standards differ across cities, even when based on the same international guidelines.

Table 1. Comparative air-quality thresholds and health-based categories used in this study, compiled based on official sources from Hong Kong, Mexico City, and the WHO (EPD, 2025a; SEMARNAT, 2023; World Health Organization, 2021).

Pollutant	Hong Kong (AQOs)	Mexico City (IAYS, NOM-172-SEMARNAT-2023)	WHO (AQGs)
PM_{10}	24h: 75 $\mu\text{g}/\text{m}^3$ (≤ 9 exceed./yr). Annual: 30 $\mu\text{g}/\text{m}^3$.	12h NowCast: Good ≤ 45 $\mu\text{g}/\text{m}^3$; Acceptable 46–60 $\mu\text{g}/\text{m}^3$ (2024) / 46–50 $\mu\text{g}/\text{m}^3$ (2026); Poor > 60 –132 $\mu\text{g}/\text{m}^3$ (2024) / > 50 –132 $\mu\text{g}/\text{m}^3$ (2026); Very Poor 133–213 $\mu\text{g}/\text{m}^3$; Extreme > 213 $\mu\text{g}/\text{m}^3$.	24h: 45 $\mu\text{g}/\text{m}^3$. Annual: 15 $\mu\text{g}/\text{m}^3$.
$PM_{2.5}$	24h: 37.5 $\mu\text{g}/\text{m}^3$ (≤ 18 exceed./yr). Annual: 15 $\mu\text{g}/\text{m}^3$.	12h NowCast: Good ≤ 15 $\mu\text{g}/\text{m}^3$; Acceptable 16–33 $\mu\text{g}/\text{m}^3$ (2024) / 16–25 $\mu\text{g}/\text{m}^3$ (2026); Poor > 33 –79 $\mu\text{g}/\text{m}^3$ (2024) / > 25 –79 $\mu\text{g}/\text{m}^3$ (2026); Very Poor 80–130 $\mu\text{g}/\text{m}^3$; Extreme > 130 $\mu\text{g}/\text{m}^3$.	24h: 15 $\mu\text{g}/\text{m}^3$. Annual: 5 $\mu\text{g}/\text{m}^3$.
O_3	8h: 160 $\mu\text{g}/\text{m}^3$ (≈ 0.08 ppm, ≤ 9 exceed./yr). Peak season: 100 $\mu\text{g}/\text{m}^3$.	1h: Good ≤ 0.058 ppm; Acceptable 0.059–0.090 ppm; Poor 0.091–0.135 ppm; Very Poor 0.136–0.175 ppm; Extreme > 0.175 ppm.	8h: 100 $\mu\text{g}/\text{m}^3$ (≈ 0.05 ppm).
NO_2	1h: 200 $\mu\text{g}/\text{m}^3$ (≈ 0.106 ppm, ≤ 18 exceed./yr). 24h: 120 $\mu\text{g}/\text{m}^3$. Annual: 40 $\mu\text{g}/\text{m}^3$.	1h: Good ≤ 0.053 ppm; Acceptable 0.054–0.106 ppm; Poor 0.107–0.160 ppm; Very Poor 0.161–0.213 ppm; Extreme > 0.213 ppm.	24h: 25 $\mu\text{g}/\text{m}^3$ (≈ 0.013 ppm). Annual: 10 $\mu\text{g}/\text{m}^3$ (≈ 0.005 ppm).
SO_2	10-min: 500 $\mu\text{g}/\text{m}^3$ (≤ 3 exceed./yr). 24h: 40 $\mu\text{g}/\text{m}^3$.	1h: Good ≤ 0.035 ppm; Acceptable 0.036–0.075 ppm; Poor 0.076–0.185 ppm; Very Poor 0.186–0.304 ppm; Extreme > 0.304 ppm.	24h: 40 $\mu\text{g}/\text{m}^3$ (≈ 0.015 ppm).

2.5 Hong Kong meteorological regimes and pollution episodes

Hong Kong’s air-pollution climatology emerges from the interaction of several regional and local meteorological processes, including winter monsoon inflow, tropical-cyclone-related subsidence, and coastal recirculation around its complex terrain and island structure (Liu & Chan, 2002; Lu & Wang, 2004; Luo et al., 2018; Zeng et al., 2025).

Luo et al. (2018) documented how northerly and north-easterly winds under the East Asian winter monsoon transport pollutants from mainland China into Hong Kong, frequently raising background concentrations during the cool season. Zeng et al. (2025) further showed that tropical cyclones modulate pollution depending on their track: eastern-track storms can enhance PM_{2.5} through subsidence and suppressed mixing, whereas western-track storms often ventilate the region by strengthening southerly flow.

On local scales, Liu & Chan (2002) and Lu & Wang (2004) highlighted how sea-land breezes, weak-pressure gradients, and stagnant episodes couple PM, O₃, and NO₂, producing multipollutant peaks when subsidence and light winds persist. These combined processes are especially influential after the passage of tropical cyclones, when clear skies, sinking motion, and low wind speeds reinforce stability and allow pollutants to accumulate across the urban areas of Hong Kong.

2.6 Comparative summary: Mexico City vs Hong Kong

When examining the physical mechanisms described in previous studies, both cities show pollution episodes linked to atmospheric conditions that limit dispersion—specifically, subsident air masses combined with low wind speeds. This general mechanism appears consistently across studies from both regions, indicating that reduced ventilation is a common “pre-requisite” for high-pollution days even though the surrounding geography and emission patterns vary significantly.

However, as outlined in previous studies, the conditions that lead to limited ventilation originate from different large-scale drivers in each city. In Mexico City, literature has identified prolonged stability associated with mid-level high-pressure systems and the seasonal structural features of the basin atmosphere (Burgos-Cuevas et al., 2021; Díaz-Esteban et al., 2022). In contrast, research on Hong Kong highlights the role of external influences—such as regional transport during the winter monsoon and stability linked to tropical-cyclone dynamics—as well as local circulation patterns that shape how multiple pollutants interact under weak-flow conditions (Liu & Chan, 2002; Lu & Wang, 2004; Luo et al., 2018; Zeng et al., 2025). Together, these studies show that while both cities experience stagnant conditions during pollution episodes, the underlying meteorological triggers arise from fundamentally different scales and processes.

3. STUDY DESIGN

This section outlines the progression of the guided study, from the initial exploratory comparisons and station selection to the consolidation of literature and, finally, the synoptic composite analysis. The project was organized into six stages—referred to as *Weeks* for consistency with the workflow used throughout the semester—and each stage contributed a specific component to the development of the research question, the data-processing pipeline, and the methodological framework.

3.1 Week 1 – Initial exploratory analysis and early framing

The project started with an exploratory analysis using January 2024 data from a small set of representative stations in both cities. For Hong Kong (HK), the stations selected were Tap Mun (rural) and Causeway Bay (urban), while in Mexico City (CDMX) the stations included Camarones (industrial), Merced (urban), and Pedregal (peripheral). The main objective at this stage was simply to familiarize with pollutant behavior, data structure, and the basic contrasts between Hong Kong and Mexico City. This initial analysis focused on four pollutants—NO₂, O₃, PM₁₀, and PM_{2.5}—and on identifying exceedances relative to local regulatory thresholds.

As summarized in **Table 2**, the first comparison showed that rural or peripheral stations tended to record higher O₃ levels, whereas urban and industrial stations displayed higher concentrations of NO₂ and particulate matter. The expected negative correlation between NO₂ and O₃ was evident in the urban stations, likely reflecting titration effects, while PM₁₀ and PM_{2.5} displayed a strong positive correlation. Although simple, these results helped confirm that the basic pollutant patterns in the selected stations aligned with established photochemical and urban-emissions behavior.

Table 2. Mean concentration for each station and pollutant (µg/m³).

Station	NO ₂	O ₃	PM ₁₀	PM _{2.5}
<i>HK Urban</i>	75.52	43.35	52.23	33.82
<i>HK Rural</i>	13.14	85.11	33.18	20.86
<i>CDMX Industrial</i>	34.50	19.33	45.08	20.98
<i>CDMX Urban</i>	38.89	18.32	43.05	21.62
<i>CDMX Peripheral</i>	25.13	29.07	29.90	14.49

This initial exploration revealed that a larger temporal window and more stations in both cities would give a better view of seasonal cycles, long-term variability, and spatial differences across networks. These insights guided the next stage of this study.

3.2 Week 2 – Full dataset assembly (2012–2024) and station classification

During *Week 2*, the analysis was expanded to cover the full 2012-2024 period, selected based on data availability from Mexico City’s air-quality site (Aire CDMX, 2025a). This time, it integrated all five major pollutants, adding SO₂ to the four previously explored, and required gathering daily records from several stations in each city. At this point, stations were grouped into four categories—urban, new town, rural, and peripheral—using information from HK’s AQHI site (AQHI, 2025) and personal familiarity with the Mexico City monitoring network.

A selection of ten stations per city was then made to ensure consistent comparisons: in the MCMA, only ten of the 38 stations had complete records for all five pollutants, and this number was matched by selecting ten of the 18 available stations in Hong Kong. Mexico City’s data required downloading, assembling, and cleaning more than one hundred individual datasets (Aire CDMX, 2025a), while Hong Kong’s platform was sufficient by downloading only two consolidated datasets containing the necessary pollutants, stations, and time periods, which were then easily combined and cleaned (EPD, 2025b). This contrast marked the first major difference in data accessibility between the two systems. The final set of ten stations for each city is summarized in **Table 3**, including their locations, altitudes, and stations types.

Table 3. Station selection and pollutant coverage, 2012-2024. Data for Hong Kong stations was obtained from official source information, while station metadata for Mexico City was obtained from an official dataset (AQHI, 2025; UPIITA/IPN, 2022)

City	Station	Lat (°N)	Lon (°E)	Alt (m)	Type	City	Station	Lat (°N)	Lon (°E)	Alt (m)	Type
HK	Central / Western	22.28	114.14	195	Urban	CDMX	Ajusco Medio	19.15	-99.16	2548	Peripheral
HK	Eastern	22.28	114.22	123	Urban	CDMX	Camarones	19.37	-99.16	2233	Urban
HK	Kwun Tong	22.31	114.23	15	Urban	CDMX	Hospital GM	19.41	-99.15	2234	Urban
HK	Sham Shui Po	22.33	114.16	14	Urban	CDMX	Merced	19.42	-99.12	2245	Urban
HK	Shatin	22.38	114.18	54	New Town	CDMX	Miguel Hidalgo	19.4	-99.2	2327	Urban
HK	Tai Po	22.45	114.17	38	New Town	CDMX	Pedregal	19.32	-99.22	2326	Peripheral
HK	Tap Mun	22.47	114.36	32	Rural	CDMX	San Agustín	19.53	-99.03	2214	Urban
HK	Tsuen Wan	22.39	114.11	35	Urban	CDMX	Tlalnepantla	19.53	-99.12	2311	Urban
HK	Tung Chung	22.29	113.94	14	New Town	CDMX	UAM Iztapalapa	19.36	-99.06	2210	Urban
HK	Yuen Long	22.45	114.16	14	New Town	CDMX	Xalostoc	19.53	-99.08	2160	Urban

After assembling the long-term dataset, basic analyses revealed clear seasonal signatures in both cities. Mexico City showed higher and more variable PM_{10} and $\text{PM}_{2.5}$, particularly during winter and spring, whereas Hong Kong displayed stronger seasonal and spatial patterns in O_3 , NO_2 , and SO_2 . For instance, O_3 peaks in autumn, while NO_2 and SO_2 show their highest concentrations in winter. Spatially, urban areas in Mexico City were most affected by particulates, and rural Hong Kong consistently displayed the highest O_3 levels. The correlations observed during *Week 1* also appeared here: PM_{10} and $\text{PM}_{2.5}$ remained strongly and positively correlated, while O_3 again showed a negative correlation with NO_2 .

These initial long-term insights indicated the need to refine the spatial categories, examine monthly patterns in greater detail, and start to incorporate existing literature on regional meteorology, pollution sources, and air-quality index design, which are part of next *Week's* work.

3.3 Week 3 – Refining station types + early conceptual understanding

In *Week 3*, the spatial classification was simplified by combining the original four station categories into two broader groups—urban and rural—to make comparisons more consistent and highlight the contrasts that mattered most for both cities. This adjustment reduced the dataset to one rural station in Hong Kong and two in Mexico City, while keeping a representative number of urban sites.

With the refined classification, a re-examination of the monthly pollutant cycles was performed, as a continuation of *Week 2*. Although the main seasonal patterns were already evident, the simplified grouping made them easier to interpret and compare across cities. The exercise confirmed that some pollutants showed clear seasonal contrasts and that spatial differences—particularly between urban and rural areas—were meaningful enough to justify focusing on these two categories in the remainder of the project, rather than retaining the initial four types.

This stage also marked the beginning of a more structured literature review centered on Mexico City's meteorology and the design of health-based indices, along with initial comparisons to Hong Kong's AQHI and WHO guidelines. The findings from this review reinforced the relevance of meteorological context and index design for interpreting pollution data, which helped outline the direction for the upcoming weeks of this study.

3.4 Week 4 – Deep Mexico City literature review

Week 4 was dedicated entirely to consolidating a literature-based framework for Mexico City, integrating findings on long-range aerosol transport, historical air-quality policies, health evidence, and both synoptic and local meteorological behavior. The aim was to organize the concepts introduced in previous weeks into a clearer structure that could guide the next steps of the project.

A detailed review of Saharan dust transport and its seasonality helped clarify how imported aerosols interact with local emissions and contribute to short-term health risks (Kutralam-Muniasamy et al., 2021; Raga et al., 2021). The analysis of Mexico City’s regulatory evolution also explained why recent frameworks adopted a 1-hour ozone metric, linking this decision to past policy discussions and epidemiological findings (Cromar et al., 2021; SEMARNAT, 2023). Additional studies on short- and long-term exposure emphasized how emergency visits and health outcomes vary depending on underlying medical conditions and population density across the MCMA (López-Feldman et al., 2021; Ugalde-Resano et al., 2022). Building on *Week 3*, the literature also connected boundary-layer behavior with broader synoptic patterns, highlighting that 500-hPa blocking and persistent stability are key precursors for many high-pollution events (Burgos-Cuevas et al., 2021; Díaz-Esteban et al., 2022; Salcido et al., 2019; Silva-Quiroz et al., 2019).

By the end of *Week 4*, these elements had come together to form a coherent conceptual foundation, enabling a clearer examination of how extreme pollution days are connected to mid-tropospheric circulation patterns. This naturally laid the groundwork for the next stage of the project, which involved developing and testing a composite-analysis approach to more effectively quantify and understand these anomalies and their underlying causes.

3.5 Week 5 – First NARR composites at 500/850 hPa

Week 5 marked the transition from exploratory and conceptual work to synoptic-scale analysis, as the first composite maps were developed using NARR 500-hPa and 850-hPa fields for the 2012–2024 period (NOAA - Physical Sciences Laboratory, 2025). Using the cleaned *Week 2* dataset, high-pollution “event days” were identified through monthly 90th-percentile thresh-

olds, after which the corresponding NARR fields were extracted. Geopotential-height anomalies were then computed, significance was tested using a t-test ($p < 0.05$), and the anomalies were plotted alongside the mean geopotential height and wind vectors.

Creating these initial maps was challenging, demanding the full capacity of the computer when handling large NARR NetCDF files, extensive reading of documentation (Cartopy contributors, 2025; xarray Developers, 2025), several hours of video tutorials, and small experiments with Python plotting libraries. The initial 500-hPa composites were created through a trial-and-error process, with separate maps for each major pollutant and incremental improvements in clarity and layout. Once these were complete, 850-hPa maps were produced to explore low-level flow during high-pollution days. Although the 850-hPa results were visually clearer, inconsistencies emerged when comparing them with the 500-hPa composites, suggesting an underlying methodological issue.

While it is still necessary to keep exploring specialized Python libraries like xarray and Cartopy for practical experience, identifying these inconsistencies was useful as it revealed that the anomaly definition had been applied incorrectly. This issue also helped clarify which parts of the workflow needed more structure and which steps required a clearer separation between data selection, climatology construction, and anomaly calculation. Realizing this prevented errors later in the project and made the logic behind the composites easier to understand. This discovery prompted a complete revision of the composite workflow in *Week 6*, using a more reliable day-of-year climatology that became the basis for creating consistent and interpretable maps in the final analysis.

3.6 Week 6 – Refined 500 hPa anomalies and HK comparison

Week 6 focused on rebuilding the composite analysis using a 31-day smoothed day-of-year climatology at 500 hPa and producing height anomalies (H') referenced to this daily baseline. This step formalized the composite workflow and allowed the results to be interpreted more consistently, while also enabling a clearer comparison with previous studies from both Mexico and Hong Kong. In parallel, several diagnostic checks were added—including aggregated counts of polluted days, wind-related statistics, and seasonal event distributions—to better contextualize the synoptic patterns identified in the maps.

The first step was to locate the error identified in *Week 5*, which became clear with the help of a reference Python code shared by a research assistant. The issue was the anomaly definition: instead of subtracting the 13-year global climatology from each event day, the correct approach was to compare each event day against a day-of-year climatology. Revising this step resolved the inconsistencies and produced coherent anomaly maps for all pollutants. In practice, this required reloading the exported NARR fields from *Week 5*, combining them with the improved Week 5 CSV to define the P_{90} threshold, applying the smoothed climatology, and then calculating H' anomalies for each event day. The refined workflow produced clearer composites, a consistent multipanel layout for the five pollutants, and auxiliary plots that highlighted how winds, magnitudes, and event frequencies varied across the 2012-2024 period. Although 850-hPa composites were not generated due to time constraints—despite having coded its daily climatology for potential future use—the 500-hPa fields were considered sufficiently representative, considering the mean elevation of Mexico City.

After generating the revised composites, the synoptic signals were compared with four Mexican studies, showing strong agreement with two—particularly regarding ridge-related circulation patterns and stability during high-pollution days—and surface-level consistency with the others. A parallel comparison was then conducted with Hong Kong literature, contrasting the mechanisms identified in Mexico City with the regional processes reported for Hong Kong. Although it would have been helpful to create Hong Kong composites to enhance the comparison, time constraints prevented this from being done. However, the available literature was sufficient to establish clear conceptual distinctions between the two cities. These comparisons collectively helped validate the composite results and ensured alignment with the broader scientific context.

The final part of *Week 6* involved reviewing how these outcomes addressed the original research questions. The discussion of health-based indices had been covered earlier in the project, and the refined composites produced in *Week 6* allowed the main question—identifying the large-scale circulation pattern that governs extreme pollution days in Mexico City and comparing it to Hong Kong’s meteorology—to be fully answered. Taken together, the six stages completed throughout the semester shaped the methodological workflow that is detailed in *Section 4* and set the foundation for the interpretive analysis presented later in the report.

4. DATA AND METHODS

4.1 Study regions and monitoring networks

The study examines two contrasting megacities—Mexico City and Hong Kong—whose geographical locations shape their air quality patterns. Surface air-quality data for both cities were obtained from their official monitoring networks and filtered to retain stations with relatively complete multi-pollutant records for 2012–2024 (Aire CDMX, 2025a; EPD, 2025b).

Mexico City lies in a high-altitude basin at roughly 2,300 m above sea level, surrounded by several mountain ridges that limit horizontal ventilation (Díaz-Esteban et al., 2022). Stations from the Automatic Atmospheric Monitoring Network (RAMA) are distributed across the Mexico City Metropolitan Area (MCMA), although only a small number are located in rural or peripheral zones, mainly toward the southern part of the basin (CAM, 2018). In contrast, Hong Kong is a coastal and highly urbanized region composed of many islands, low peaks, and narrow land–sea corridors. Its monitoring network covers most of the territory, with a larger proportion of urban stations and a few rural or semi-rural sites near country parks (AQHI, 2025). Despite clear differences in topography and setting, both networks provide regular measurements of the same five criteria pollutants used in this study. A summary of the station locations used in the analysis is shown in **Figure 2**, which highlights the spatial distribution of urban and rural sites across both cities.

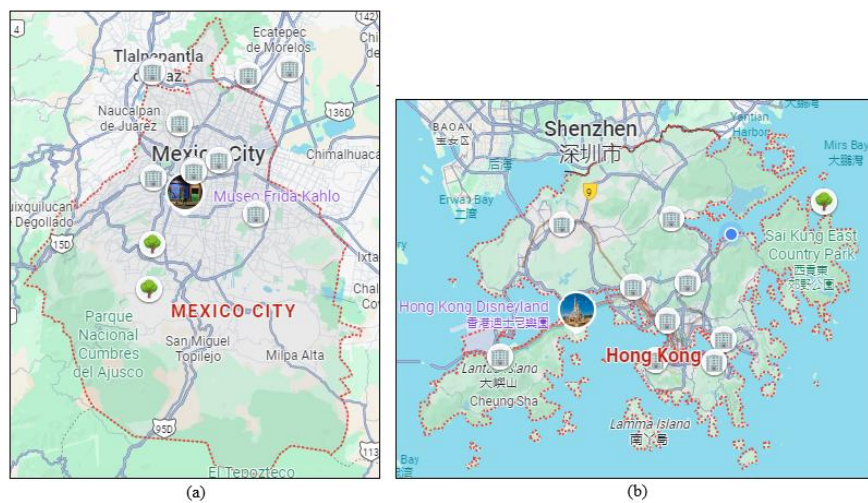


Figure 2. Spatial distribution of the monitoring stations used in this study. (a) MCMA stations; (b) Hong Kong stations. Urban stations (building icon) and rural stations (tree icon) are marked for visual clarity. Station coordinates and metadata were obtained from UPHITA/IPN dataset and AQHI/EPD official sources.

4.2 Air-quality datasets (2012–2024)

Daily surface pollutant data were obtained for both cities for the period 2012–2024, focusing on five key pollutants—PM₁₀, PM_{2.5}, O₃, NO₂ and SO₂—which are central to their respective regulatory frameworks and health-based indices (Aire CDMX, 2025b; EPD, 2025a)

For Mexico City, data availability constrained the analysis to 2012 onward (Aire CDMX, 2025a). Monthly files containing daily means for the selected stations (**Table 3**) and pollutants were downloaded individually and then concatenated into a single dataset. These files required substantial cleaning and restructuring, including handling missing values, standardizing date formats, and harmonizing pollutant names. The resulting dataset contained one row per station per day, with columns for date, station name, and pollutant concentrations.

For Hong Kong, although daily data extend back to 1991, the same 2012–2024 period was extracted to match Mexico City. The Environmental Protection Department’s platform allows selecting stations, pollutants and averaging periods prior to download, which simplified the process (EPD, 2025b). Two files were needed due to the 10-year download limit, and these were later merged and reformatted to align with the structure of the Mexico City dataset. After cleaning, each city’s dataset was kept separate but formatted in the same way, with identical column names and structure, so the analysis could be applied consistently.

4.3 NARR reanalysis data and spatial domain

NARR (North American Regional Reanalysis) was used to characterize the synoptic environment associated with high-pollution event days. Daily pressure-level fields from the NOAA archive (NOAA - Physical Sciences Laboratory, 2025) were extracted, with a primary focus on 500-hPa geopotential height and horizontal wind components. These variables were chosen because 500-hPa flow provides information about mid-tropospheric stability and large-scale circulation patterns that influence surface stagnation in the Mexico City basin.

The spatial domain for all composites was set to cover a broad region of Mexico, approximately from 12°N to 33°N and 85°W to 120°W. Using a larger domain ensured that ridges, troughs and regional wind structures were visible and not truncated near the edges of the plot,

which made the anomaly patterns easier to interpret. Within this domain, a smaller box delimiting slightly more than the Mexico City Metropolitan Area (MCMA) was added for reference, spanning roughly 18°N - 21°N and 97°W - 101°W , with a star marking the location of downtown Mexico City at 19.43°N , 99.13°W . The selected study domain with the MCMA reference box and Mexico City's star are illustrated in **Figure 3**.



Figure 3. Spatial domain used for NARR extraction. The large box shows the regional domain, and the smaller orange box outlines the Mexico City Metropolitan Area. The yellow star marks downtown Mexico City.

NARR was selected because it provides continuous, high-resolution reanalysis data for North America, which includes all of Mexico. Its temporal coverage and spatial resolution made it suitable for detecting synoptic-scale patterns during extreme pollution events, and its consistent structure allowed seamless extraction of daily fields corresponding to event dates.

4.4 Defining event days (p90 method)

High-pollution “event days” were defined using a monthly 90th-percentile threshold (p_{90}) of daily mean concentrations for each pollutant. This approach isolates roughly the top 10% of days while still providing enough samples to produce meaningful composites. Using a percentile-based threshold, rather than AQI categories, avoids misclassifying winter-spring peaks as “extreme” simply because they occur in seasons with naturally higher concentrations, and makes the definition consistent across pollutants, naturally accounting for the strong seasonality in both cities, since each month has its own p_{90} value.

Because the pollutant distributions are right-skewed, the number of event days depends on how many observations exceed each pollutant's p_{90} threshold over the 2012-2024 period. Although the exact thresholds differ among pollutants, the total number of event days remains broadly comparable across them, which ensures a similar sample size when producing the composites. The distributions and their corresponding p_{90} cutoffs are shown in **Figure 4**, where the red dashed line marks the threshold used to define the event-day sets.

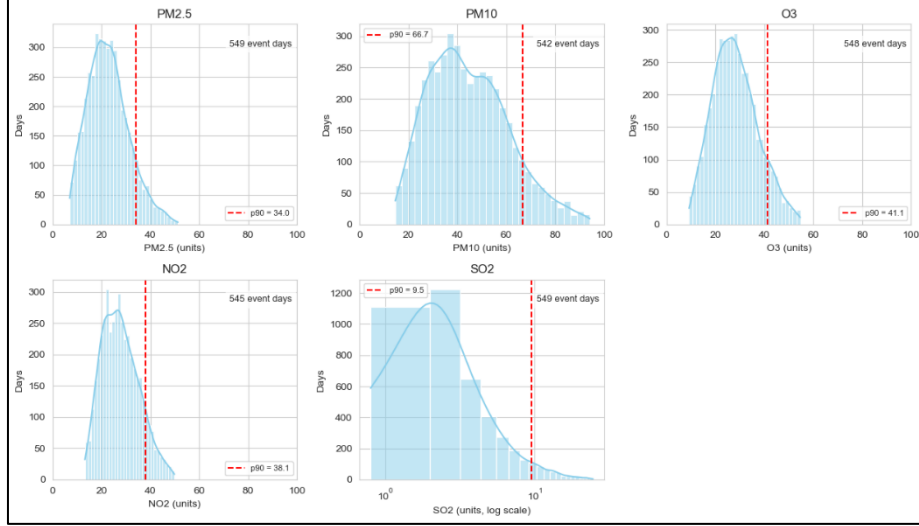


Figure 4. Daily pollutant distributions (2012-2024) with 90th-percentile thresholds (p_{90} , red dashed line) defining high-pollution event days used in the composite analysis, with uniform and standardized x-axis scale for $PM_{2.5}$, PM_{10} , O_3 & NO_2 , and logarithmic scale for SO_2 .

Since thresholds were computed independently for each pollutant, the resulting event-day sets are not identical; some days trigger only one pollutant (single-component events), while others exceed multiple thresholds (multicomponent events). These differences are handled later in the composite analysis by creating separate maps for each pollutant, which do not influence the overall composite workflow.

4.5 Day-of-year climatology for anomaly calculation

To compute anomalies consistently across seasons, a 31-day smoothed day-of-year (DOY) climatology was constructed for each NARR variable. In this context, the climatology represents the typical value of a field (e.g., geopotential height or winds) for each day of the year, averaged over all years (2012-2024). A 31-day rolling window was applied to smooth the annual cycle and reduce sharp transitions between adjacent days, which helps stabilize the baseline used in the anomaly calculation.

Using a smoothed DOY climatology guarantees that anomalies represent significant deviations from the expected seasonal pattern rather than artifacts of the raw daily data variability. Once the climatology was established, anomalies were calculated as the difference between the NARR value on each event day and the corresponding climatological value for that day of the year. This method provided a seasonally consistent measure of mid-tropospheric deviations associated with high-pollution events, ensuring that anomalous values during event days were comparable across all seasons, which is an essential factor when composites include dates from different months.

4.6 Composite workflow

The composite workflow consisted of six main steps: (1) identifying event days using the monthly p_{90} thresholds; (2) extracting the corresponding NARR fields; (3) computing the smoothed day-of-year climatology; (4) calculating height anomalies (H') for each event day; (5) assessing statistical significance using a t-test ($p < 0.05$); and (6) plotting the H' composites as shaded fields with contour lines and wind vectors over a basemap of Mexico.

Anomalies were plotted directly, without including the absolute geopotential height field, to avoid overdoing and keep the focus on deviations from seasonal conditions. Positive anomalies were represented with solid-line contours and warm colors, while negative anomalies were shown with dashed contours and cool colors, using white around zero to center the scale. This approach maintained the maps' visual clarity and consistency across pollutants.

To align the daily climatology with each event date, the dataset was grouped by day of year before applying the rolling 31-day window. For the significance analysis, stippling was used to highlight grid points where anomalies differed from climatology at $p < 0.05$, and a thinning factor was applied to avoid excessive marking. Mean horizontal winds were added as quiver arrows to provide basic flow context.

Each final map included the MCMA reference box, a star marking downtown Mexico City, labelled gridlines, and a consistent color scale. A multipanel figure was then produced using the same color limits for all pollutants, which allowed for direct side-by-side comparison. This unified output serves as the basis for the synoptic interpretation presented in *Section 5*.

4.7 Diagnostic metrics (winds, frequencies, aggregates)

Several diagnostic metrics were computed to complement the anomaly fields. These included the mean wind speed and the prevailing wind direction on event days, calculated from the horizontal wind components at 500 hPa. A summary of the mean anomaly magnitudes was also obtained to quantify the typical mid-tropospheric deviations during high-pollution days throughout the 2012-2024 period. For clarity, wind directions in *Section 5* are described in their standard meteorological convention (direction from which the flow originates), even though the internal azimuth calculations used in the vector-mean procedure refer to the direction toward which the flow points.

Additionally, a monthly event-distribution metric was created to indicate how frequently event days occurred in each month for each pollutant. This offered an easy way to visualize the seasonal patterns of events. Together, these diagnostics provided extra context for the composite maps and helped organize the later interpretation of seasonal and synoptic patterns.

4.8 Limitations

This methodological framework is subject to several limitations that should be taken into account when interpreting the results. The composite-analysis workflow was developed iteratively throughout the project, and the final implementation still faces constraints related to data availability, computational resources, and reliance on a single reanalysis product. These limitations reflect not only practical restrictions but also areas for future improvement.

A first limitation relates to the scope of the composite analysis itself. Due to time and computing-capacity constraints, only the 500-hPa composites were produced in their final form. While NARR provides multiple pressure levels, extracting, processing and storing the full set of fields demanded substantial computational resources. As a result, 850-hPa composites, which could have provided valuable information about low-level ventilation and basin-scale flow, were not completed. Similarly, relying solely on NARR means the analysis is based on one reanalysis system; using an additional one could have strengthened the robustness of the findings.

Data availability also posed limitations. For Mexico City, the full 2012–2024 period was only partially complete across stations, resulting in an uneven distribution of rural and urban sites. Rural coverage is particularly sparse in the MCMA, with only a few stations located outside the urban core. In Hong Kong, the monitoring network is more extensive, but most available stations are urban, with only one truly rural site selected (Tap Mun). Roadside stations were excluded to reduce traffic-driven biases, but doing so limited spatial coverage in some districts. These imbalances may influence how representative the event-day sets are for each of these two cities.

Finally, the implementation process itself introduced constraints. Prior to this project, working with specialized Python libraries—such as `xarray` and `Cartopy`—was unfamiliar, and this resulted in a trial-and-error workflow that slowed progress. Several steps, such as testing anomaly definitions, aligning the climatology, and building plotting functions, required substantial debugging. The composite workflow improved significantly after reviewing shared example code, but a more developed technical background would likely have reduced the overall effort and allowed the analysis to be extended to additional variables and levels.

In summary, all the procedures outlined in this section establish a coherent and reproducible workflow that connects surface pollution data with large-scale atmospheric circulation. The approach is deliberately designed to be modular: the air-quality dataset from *Week 2* forms the basis for defining event days, then the day-of-year climatology provides a seasonally consistent reference, while the anomaly calculations highlight deviations or departures from typical mid-tropospheric conditions, which leads to a final diagnosis supported by several metrics that provide further temporal and dynamic context. Although the implementation required several iterations—partly due to data constraints, partly due to technical limitations—the final workflow is sufficiently robust to support a synoptic and comprehensive assessment of extreme pollution days. More importantly, the pipeline also allows the results to be compared with the reviewed literature, both for Mexico City and Hong Kong. Therefore, with the methodological foundation completely established, the next section shows the outcomes of this thorough analysis, focusing on the composite maps, the distribution of event days, and the accompanying diagnostics that helped reveal the prevailing circulation patterns associated with high-pollution episodes in the Mexico City Metropolitan Area (MCMA).

5. RESULTS

5.1 Summary of event days and seasonal distribution

Before examining the anomaly fields, the event-day sets in Mexico City were summarized to characterize their frequency and seasonal distribution across the 2012-2024 record. As shown in **Figure 4**, the p_{90} thresholds identify approximately 540-550 event days per pollutant, with pollutant-specific cutoffs of around $34 \mu\text{g m}^{-3}$ for $\text{PM}_{2.5}$, $66 \mu\text{g m}^{-3}$ for PM_{10} , $41 \mu\text{g m}^{-3}$ for O_3 , $38 \mu\text{g m}^{-3}$ for NO_2 and $9 \mu\text{g m}^{-3}$ for SO_2 .

Figure 5 illustrates the monthly average concentrations of $\text{PM}_{2.5}$, PM_{10} , O_3 , NO_2 , and SO_2 throughout the years 2012 to 2024, offering a comprehensive overview of the annual cycle of air quality in the Mexico City Metropolitan Area. The pollutants show distinct seasonal trends: while PM_{10} and NO_2 reach their peak levels in winter and early spring, O_3 peaks in April-May, $\text{PM}_{2.5}$ has higher concentrations in winter and again in late spring, and SO_2 remains low throughout the year with a slight increase in winter. These patterns are useful for gaining a better understanding of the synoptic composites that will be discussed later.

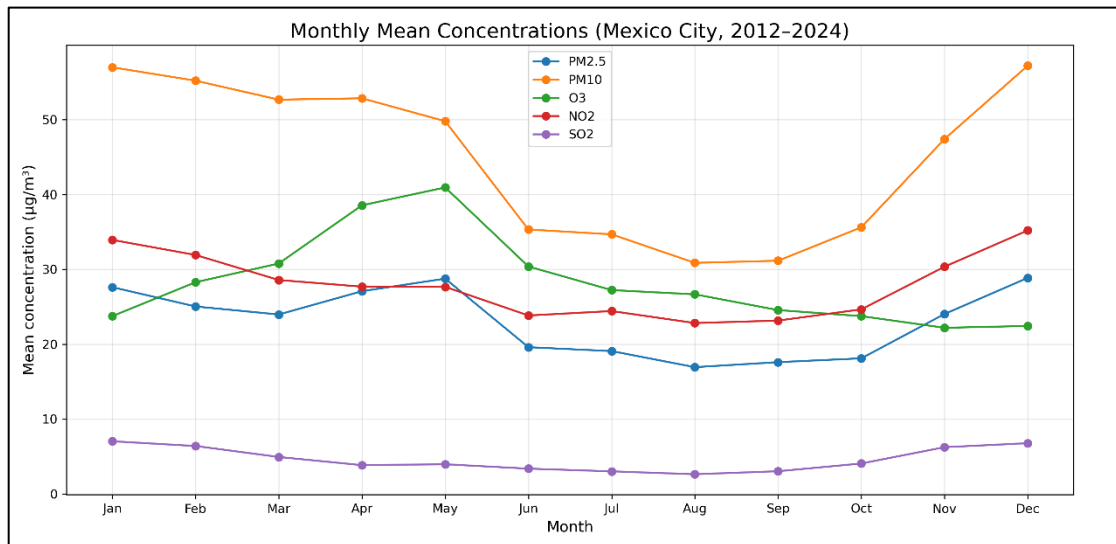


Figure 5. Monthly mean concentrations of $\text{PM}_{2.5}$, PM_{10} , O_3 , NO_2 and SO_2 in the Mexico City Metropolitan Area (2012-2024). The curves show the annual cycle of each pollutant based on daily means aggregated by month.

5.2 500-hPa anomaly composites (single-pollutant)

Composite anomaly fields at 500 hPa were computed separately for each pollutant to describe the synoptic patterns associated with high-pollution event days. Each composite shows the spatial distribution of mid-tropospheric height anomalies (H') and the corresponding mean horizontal winds for the event-day set of that pollutant. The following subsections introduce these fields individually, starting with $PM_{2.5}$ and PM_{10} , then O_3 , NO_2 , and SO_2 . Although all composites share the same spatial domain and plotting conventions, each pollutant shows differences in anomaly amplitude and spatial coherence, which are summarized below.

5.2.1 $PM_{2.5}$

The $PM_{2.5}$ composite shows a broad and coherent region of positive height anomalies centered over central Mexico (**Figure 6**). The anomaly maximum reaches values of approximately +4 m and extends across the central plateau with a smooth northeast-southwest orientation. Mean 500-hPa winds over the MCMA region are weak and predominantly easterly to east-northeasterly, with typical speeds of about 1-2 $m\ s^{-1}$. Farther north in the domain, the flow becomes westerly and stronger, but these features represent regional circulation outside the basin. Overall, the composite presents a spatially extensive positive anomaly with limited areas of negative values and relatively uniform weak flow over central Mexico.

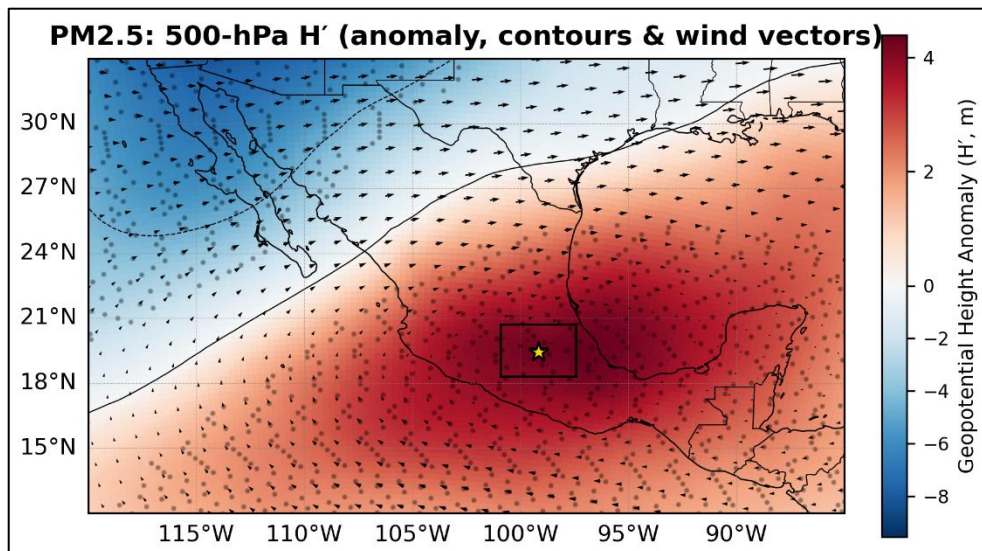


Figure 6. $PM_{2.5}$ composite at 500 hPa showing geopotential-height anomalies H' (shading, m), H' contours (solid = positive; dashed = negative) and mean 500-hPa wind vectors for $PM_{2.5}$ event days (2012-2024). The MCMA box and Mexico City marker are included for reference.

5.2.2 PM₁₀

The PM₁₀ composite displays a broadly similar anomaly structure to that of PM_{2.5}, but with a larger amplitude and more spatially extensive positive values (**Figure 7**). The anomaly maximum reaches approximately +6 m and covers much of the central plateau, extending eastward toward the Gulf of Mexico. The positive region is smooth and continuous, with closed contours surrounding the anomaly core.

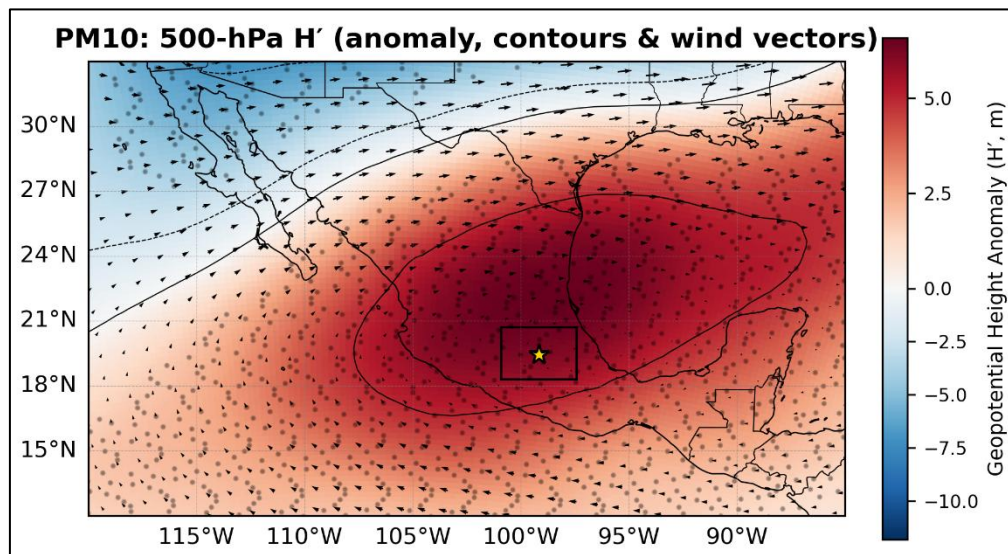


Figure 7. PM10 composite at 500 hPa showing geopotential-height anomalies H' (shading, m), H' contours (solid = positive; dashed = negative) and mean 500-hPa wind vectors for PM10 event days (2012-2024). The MCMA box and Mexico City marker are included for reference.

Mean 500-hPa winds over the MCMA region are very weak and predominantly easterly to east-northeasterly, with typical speeds even below 1 m s^{-1} . Farther north in the domain, the flow becomes westerly and stronger, but these features occur well outside the basin. Overall, the PM₁₀ composite exhibits a widespread positive anomaly field with only limited negative values near the far northwestern edge of the domain and relatively uniform weak flow over central Mexico.

5.2.3 O₃

The O₃ composite shows a more spatially concentrated anomaly field compared with the particulate pollutants, with a clearly defined positive core over central Mexico (**Figure 8**). The maximum anomaly reaches approximately +5-6 m and extends northwestward from the central

plateau toward the Pacific side of the country. The positive region is smooth and fairly continuous, covering most of the Mexican domain, while the MCMA lies along the southeastern edge of the anomaly core.

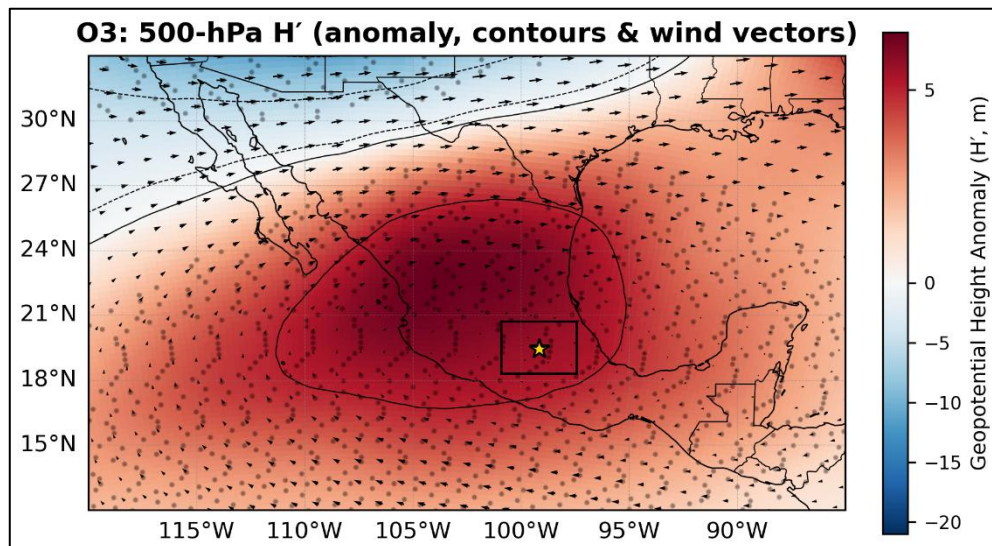


Figure 8. O₃ composite at 500 hPa showing geopotential-height anomalies H' (shading, m), H' contours (solid = positive; dashed = negative), and mean 500-hPa wind vectors for O₃ event days (2012-2024). The MCMA box and Mexico City marker are included for reference.

Mean 500-hPa winds over the MCMA region are the weakest among all pollutants and are predominantly easterly to east-northeasterly, with typical speeds of around 0.75 m s^{-1} . Farther north, the flow becomes stronger and westerly, but these features occur outside the basin and do not affect the central-plateau region. Overall, the O₃ composite exhibits a widespread positive anomaly field with only small negative areas in the far northwest and generally uniform weak flow across central Mexico.

5.2.4 NO₂

The NO₂ composite shows a weaker but still well-defined anomaly pattern, with a clear positive core centered over the central plateau (**Figure 9**). The anomaly maximum reaches approximately +4-5 m and extends northward across much of the domain. The positive region is smooth and continuous, covering most of Mexico, while the MCMA lies within the central portion of the anomaly core.

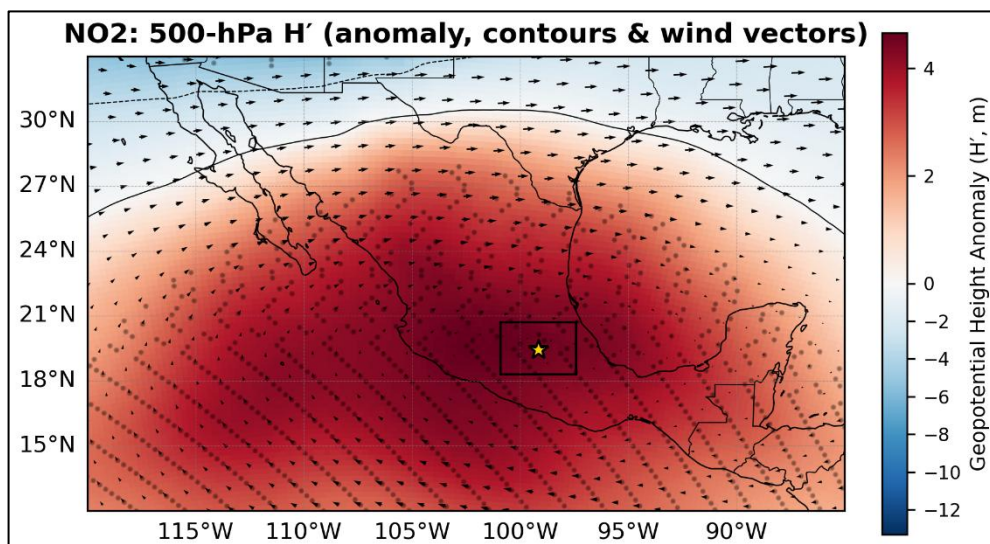


Figure 9. NO₂ composite at 500 hPa showing geopotential-height anomalies H' (shading, m), H' contours (solid = positive; dashed = negative), and mean 500-hPa wind vectors for NO₂ event days (2012-2024). The MCMA box and Mexico City marker are included for reference.

Mean 500-hPa winds over the MCMA region are weak and predominantly easterly to east-northeasterly, with typical speeds of around 1.3 m s^{-1} . Farther northwest, the flow becomes stronger and westerly, but these features occur outside the basin and do not influence the central region. Overall, the NO₂ composite exhibits a widespread positive anomaly field with a broad, rounded structure and generally uniform weak flow over central Mexico.

5.2.5 SO₂

The SO₂ composite shows the weakest anomaly amplitudes among all pollutants, although the spatial pattern remains broadly like the others (**Figure 10**). The positive anomalies over central Mexico reach approximately +4-5 m, with the highest values confined to the far northern edge of the domain, outside the Mexican region. The anomaly field is smooth and continuous over most of the country, with the MCMA lying within a modest positive area.

Mean 500-hPa winds over the MCMA region are weak and predominantly easterly to east-northeasterly, with typical speeds below 0.8 m s^{-1} . Farther north, the flow becomes westerly and stronger, but these features occur outside the basin and do not influence the central plateau. Overall, the SO₂ composite exhibits a broad, low-amplitude positive anomaly field with a consistent weak-flow environment over central Mexico and only small negative areas along the far northeastern boundary of the domain.

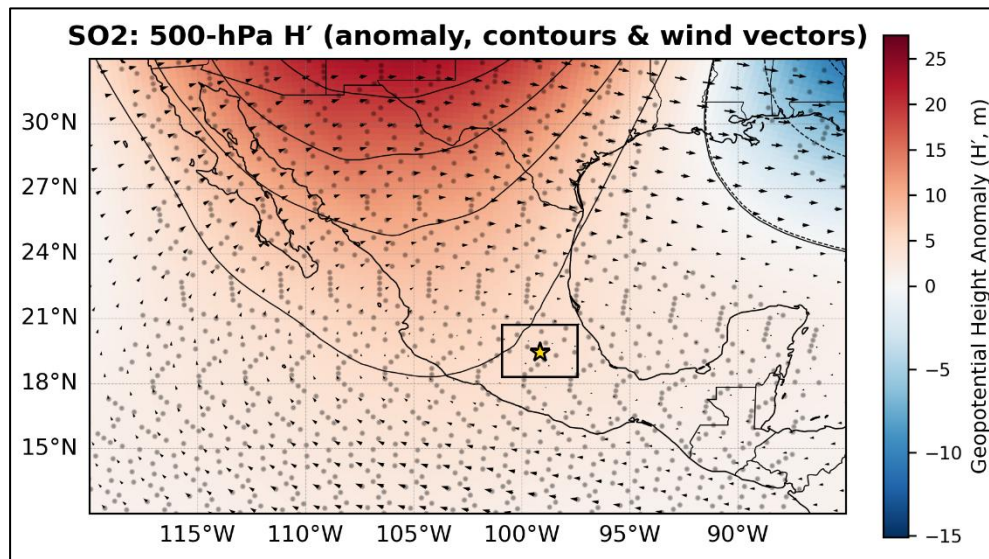


Figure 10. SO₂ composite at 500 hPa showing geopotential-height anomalies H' (shading, m), H' contours (solid = positive; dashed = negative), and mean 500-hPa wind vectors for SO₂ event days (2012-2024). The MCMA box and Mexico City marker are included for reference.

5.3 Combined multipollutant composite comparison

To enable direct comparison across pollutants, a unified multipanel figure was created using identical color limits, spatial boundaries and plotting conventions (**Figure 11**). When placed side by side, the five composites reveal several visual similarities, particularly the presence of positive anomalies over central Mexico and the broadly consistent orientation of the patterns. All panels show a dominance of positive height anomalies over the MCMA region, with only small negative areas appearing near the far northern or northwestern edges of the domain. The wind fields are also weak across all composites, with easterly to east-northeasterly flow over central Mexico.

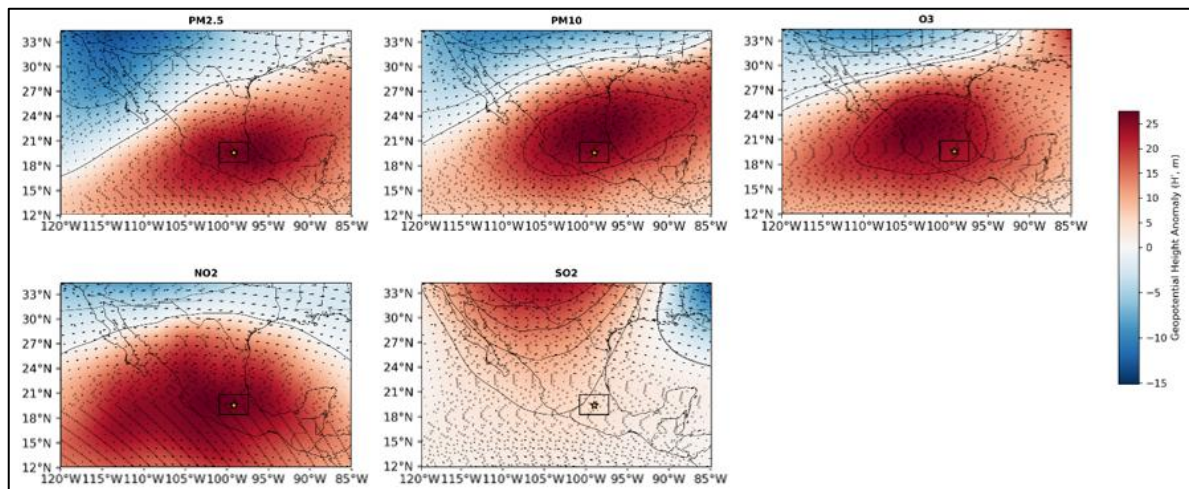


Figure 11. Multipollutant 500-hPa anomaly composites (H') for PM_{2.5}, PM₁₀, O₃, NO₂, and SO₂ using consistent color limits and plotting conventions. The MCMA box and Mexico City marker are included for reference.

Despite these common elements, the composites differ in amplitude and spatial coherence. PM₁₀ and O₃ exhibit more compact positive cores, with PM₁₀ extending slightly toward the Gulf of Mexico and O₃ showing a northwestward projection. PM_{2.5} and NO₂ present broader, more diffuse structures that cover a large portion of the central plateau, while SO₂ shows the weakest anomalies but retains the same positive pattern. Together, the multipanel comparison highlights both the shared and pollutant-specific characteristics of the anomaly fields.

5.4 Diagnostic metrics (winds, magnitudes, aggregates)

Additional diagnostics were computed to complement the composite maps and provide a simple quantitative summary of the event-day characteristics. **Table 4** summarizes the mean anomaly magnitudes and 500-hPa wind characteristics for each pollutant. Mean H' values range from roughly 4 to 6 m, with PM₁₀ showing the strongest average anomaly and PM_{2.5} the weakest, while O₃ and NO₂ fall in between. Wind speeds remain consistently low across all pollutants—generally below 2 m s⁻¹—reflecting the weak-flow environment observed in the composite maps. Predominant winds come from easterly to east-northeasterly directions, consistent across pollutants and matching the weak, basin-top circulation seen during event days.

Table 4. Mean 500-hPa anomaly magnitude (H'), average wind speed, and predominant wind direction for event-day composites of PM_{2.5}, PM₁₀, O₃, NO₂ and SO₂ (2012–2024). Wind directions are expressed in the standard meteorological convention (direction from which the flow originates).

Pollutant	Mean H' (m)	Mean wind speed (m s ⁻¹)	Predominant wind direction (from)
PM _{2.5}	4.17	1.7	E–ENE
PM ₁₀	6.1	0.91	E–NE
O ₃	5.58	0.75	E–ENE
NO ₂	4.51	1.35	E–ENE
SO ₂	4.44	0.78	E–ENE

Event-day frequencies, aggregated by month reinforce the seasonal signatures identified earlier. PM_{2.5} shows two concentration periods, one in winter and another in late spring, while PM₁₀ and NO₂ have maximum levels in winter and early spring. O₃, on the other hand, peaks only in April–May, whereas SO₂ levels stay low all year, with only a slight rise in winter.. Together, these diagnostics provide a statistical summary of the event-day environment and complement the spatial composites presented in previous sections.

6. DISCUSSION

This section interprets the synoptic composites and diagnostic metrics presented in *Section 5*, connecting them with the literature reviewed in *Section 2*. The aim is to situate the empirical findings within a broader meteorological and air-quality context, highlight the physical processes that govern extreme-pollution days in Mexico City, and compare them with the well-documented regimes affecting Hong Kong. Rather than repeating the previous results, the discussion focuses on how these patterns answer the main research question outlined in *Section 1* and what they imply for understanding and managing air pollution in both cities.

6.1 Dominant synoptic pattern controlling pollution peaks in Mexico City

Across pollutants, the composites consistently show that high-pollution days in Mexico City occur under a mid-tropospheric ridge, characterized by positive H' anomalies and weak easterly to east-northeasterly winds over the basin. This recurrent pattern appears in all five pollutants despite differences in spatial extent and amplitude, pointing to a shared synoptic forcing behind extreme events. On the maps, the ridge manifests as smoothly varying positive anomalies with reddish shading and closed contours over central Mexico, while the associated weak winds indicate limited horizontal ventilation over the plateau. This provides a clear synoptic ‘fingerprint’ for extreme-pollution days in the MCMA.

The physical mechanism behind this ridge-driven stagnation aligns with boundary-layer studies that describe shallow daytime mixing and stable nocturnal conditions during the dry season. Subsidence associated with the ridge warms the mid-troposphere, strengthens capping inversions, and suppresses vertical mixing, favoring pollutant accumulation near the surface. This is consistent with the stability patterns described by Díaz-Esteban et al. (2022), the weak upper-level winds and poor ventilation identified by Salcido et al. (2019), and the association between 500-hPa blocking and high-ozone contingency days documented by Silva-Quiroz et al. (2019). Earlier work also pointed to limited 500-hPa control and strong local recirculation within the basin, which is compatible with the ridge-stagnation regime seen in the composites (Jauregui Ostos & Luyando, 1992).

Conceptually, the ridge acts as a warm “lid” that caps the boundary layer: a shallow mixing depth and frequent inversions limit the height of the mixed layer, reduce ventilation, and allow

pollutants to build up. This picture matches common observations in the MCMA, such as recurrent thermal inversions in winter and spring and a more stable ABL during the dry season. Urban heat and pollution islands further contribute to this stability and spatial variability, reinforcing hotspots in central areas compared with surrounding rural zones (Burgos-Cuevas et al., 2021; Mendez-Astudillo et al., 2022).

The seasonal cycle derived from the monthly mean concentrations (**Figure 5**) supports this synoptic interpretation. O_3 peaks in April-May, when strong radiation and weak ventilation favor photochemical production prior to the rainy season. PM_{10} and NO_2 reach their highest levels in winter and early spring under shallow boundary-layer conditions, while $PM_{2.5}$ shows a semi-bimodal pattern with elevated values in winter and again in late spring. SO_2 remains low throughout the year with a slight enhancement during the colder months. These seasonal features are consistent with the ridge-stagnation mechanism identified in the composites, illustrating how stability, radiation, and reduced mixing jointly modulate pollutant levels across the basin. Overall, these cycles reflect the combined influence of the dry-season stability, temperature changes, and the transition into the rainy season. Months with clear skies and weak winds allow pollutants to accumulate more easily, while convective activity and deeper mixing during summer dilute concentrations.

6.2 Differences among pollutants and what they reveal

Although all pollutants share the same synoptic background, their anomaly fields differ in amplitude and spatial expression. $PM_{2.5}$ and NO_2 display broad, spatially widespread positive anomalies that extend over most of central Mexico, consistent with their behavior under weak-flow conditions and their tendency to accumulate when ventilation is limited. In contrast, PM_{10} and O_3 retain broadly similar structures but exhibit more spatially defined anomaly cores, with PM_{10} showing the highest overall anomaly magnitude and O_3 displaying a slightly more concentrated center over the central plateau. The contrast between $PM_{2.5}$ and PM_{10} is particularly informative. PM_{10} presents the strongest anomalies—consistent with its sensitivity to surface resuspension and stability—while $PM_{2.5}$ shows a more uniform distribution and a smaller anomaly contrast, reflecting its mixture of urban and rural contributions. This difference is also visible in their seasonal curves, where PM_{10} somewhat resembles $PM_{2.5}$ during winter, but then they diverge in the spring months.

The SO₂ composite displays the weakest anomaly field of all pollutants, with only a modest positive signal over central Mexico and large areas of near-zero anomalies. This limited spatial expression aligns with SO₂'s short lifetime and localized emission sources. The weak-flow environment during SO₂ event days is consistent with the low-intensity patterns reported by Salcido et al. (2019) and is further supported by **Figure 5**, which suggests that SO₂ responds to the same synoptic stability that shapes other pollutants, albeit with a reduced spatial footprint.

6.3 Comparison with previous literature (Mexico City)

The ridge signature identified in the composites aligns closely with previous studies linking Mexico City's worst pollution episodes to mid-tropospheric blocking. Silva-Quiroz et al. (2019) associated ozone contingency days with weak-flow 500-hPa highs, while Díaz-Esteban et al. (2022) classified long-term circulation into seven recurrent patterns, showing that ridge-like systems dominate O₃ peaks in spring and particulate peaks in winter. The positive mid-tropospheric anomalies obtained here fall within the expected range reported in those studies and reproduce the characteristic blocking and stagnation regimes observed during high-pollution events.

Local-scale processes documented in ABL-focused studies also support this interpretation. Burgos-Cuevas et al. (2021) and Méndez-Astudillo et al. (2022) describe multilayered inversions, nocturnal convergence, and interactions between urban heat and pollution islands that reinforce stability within the basin. These mechanisms explain why the ridge pattern is so effective at producing extreme events: synoptic subsidence limits vertical mixing, while local stability features further reduce ventilation and amplify spatial gradients across the city. Together, these results show strong consistency between the composites and previously described meteorological controls on pollution in the MCMA.

6.4 Comparative context and implications

Mexico City and Hong Kong provide two contrasting yet meteorologically comparable frameworks for interpreting the ridge-stagnation mechanism identified in this study. Both cities experience high-pollution episodes under weak-wind, subsident conditions, but their origins differ fundamentally. In Mexico City, peaks arise from locally driven stagnation beneath a mid-tropospheric ridge, which suppresses boundary-layer development and limits ventilation. In

contrast, Hong Kong's most severe episodes are largely imported, shaped by the East Asian winter monsoon and by subsidence linked to distant tropical cyclones (Luo et al., 2018; Zeng et al., 2025). Its coastal-island topography also facilitates sea-land breeze recirculation, generating daytime convergence and nocturnal stability that trap pollutants locally (Liu & Chan, 2002), while multi-pollutant coupling strengthens under post-cyclone subsidence (Lu & Wang, 2004). Despite similar latitudes and population densities, the synoptic forcing differs sharply: Mexico City's events are internally generated, Hong Kong's externally modulated.

These contrasting mechanisms imply distinct forecasting and policy strategies. In Mexico City, early warnings depend on monitoring developing ridges, stability indicators, and weak mid-tropospheric flow. In Hong Kong, communication relies more on tracking regional transport pathways, monsoon phases, and tropical-cyclone positions. This meteorological divergence helps explain why each city employs different health-risk metrics: Mexico City's IAYS uses pollutant-specific thresholds and 1-hour ozone values, whereas Hong Kong's AQHI applies a 3-hour multi-pollutant formulation based on hospital-risk weighting (Environmental Protection Department, 2025; SEMARNAT, 2023). Hong Kong's system benefits from predictable synoptic precursors—often 2 to 3 days in advance—while Mexico City requires real-time monitoring of boundary-layer structure and incoming ridges.

Taken together, the results show that Mexico City's extreme pollution days are driven by internally produced stagnation under a recurring ridge-based synoptic pattern, while Hong Kong's are shaped by regionally forced circulation. These contrasts clarify why pollutants behave differently in each city and emphasize the need for region-specific forecasting tools and public-health communication strategies. In doing so, this section also answers the research question by placing the composites within their wider meteorological context and demonstrating how differences between cities influence air-quality outcomes. Additionally, these meteorological differences emphasize the important lesson that air-quality management must be tailored to each city. Even with similar pollutants and latitudes, the main atmospheric factors, health risk profiles, and policy requirements can vary significantly.

7. CONCLUSIONS

This guided study examined the large-scale meteorological conditions linked to extreme pollution days in Mexico City and compared them conceptually with those in Hong Kong's regional regimes. Through six analysis stages—using surface records, NARR reanalysis, and monthly climatology—the findings identified a dominant synoptic pattern influencing high-pollution days in Mexico City. The composite workflow and data exploration fully addressed the initial research question, clarifying how these events form and persist.

Across all pollutants, synoptic analyses revealed a recurring mid-tropospheric ridge with weak easterly flow and limited ventilation over the basin. This ridge-stagnation mechanism promotes subsidence, strengthens thermal inversions, and suppresses vertical mixing, trapping pollutants near the surface. The seasonal cycle in **Figure 5** supports this, showing that winter, early spring, and the late-spring stability window favor pollutant buildup, despite each pollutant's seasonal behavior. These patterns align with previous Mexican studies linking stagnation, inversions, and weak flow to high-pollution days.

In contrast, Hong Kong's extreme pollution episodes arise from regional-scale processes, including monsoon-driven northeasterlies, sea-land breeze interactions, and subsidence associated with tropical cyclones. Although both cities experience pollution during weak-wind, subsident conditions, the underlying causes differ: Mexico City's peaks are largely home-grown within an enclosed basin, while Hong Kong's are often imported or modulated by regional circulation. These differences illustrate why each city employs its own air-quality methods, which are shaped by local meteorology as well as institutional and policy choices.

Taken together, these findings provide a clear picture of the synoptic environment that governs Mexico City's extreme pollution days and outline several directions for potential future work, which could include repeating the composite analysis at different pressure levels, developing Hong Kong-specific composites, refining event-day thresholds, and examining interannual variability such as ENSO years. Beyond this guided study, the methods and results developed here could serve as the basis for a thesis or a future publication. Overall, the study successfully met its objectives and shows that synoptic-scale analysis can meaningfully support both air-quality forecasting and public-health communication in complex urban environments.

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